RouteRec: Behavior-Guided Expert Routing for Sequential Recommendation

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Abstract

Sequential recommendation exhibits heterogeneous behavioral regimes, yet most models still process all prefixes through largely shared computation. In practice, some prefixes are governed by stable preference, others by evolving session intent, and others by immediate local transitions. This creates a mismatch between behavioral diversity and shared computation. Mixture-of-experts (MoE) offers a natural mechanism for conditional computation, but hidden-state-driven routing remains limited for this setting: it is often semantically opaque and does not explicitly align expert allocation with different temporal scopes.

We propose **RouteRec**, a behavior-guided MoE framework for sequential recommendation built around three principles. First, RouteRec derives *portable behavioral cues* from ordinary interaction logs, avoiding dependence on heavy side-information encoders. Second, it applies *coarse-to-fine multi-stage routing* across macro, mid, and micro scopes so that long-range tendency, within-session evolution, and immediate transition cues can influence routing at different granularities. Third, it uses *behavior-guided expert allocation* that decouples routing signals from hidden representations, making expert selection more inspectable and behavior-grounded.

This design treats behavior features as routing controls rather than as yet another representation-fusion channel. As a result, RouteRec preserves strong sequence modeling in the representation path while introducing a lightweight and transferable control interface for conditional computation. Experiments show that RouteRec improves ranking quality on several datasets while remaining competitive on others, and yields more behavior-consistent routing patterns in diagnostic analyses.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

sequential recommendation, mixture of experts, conditional computation, routing, behavioral features

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2026.

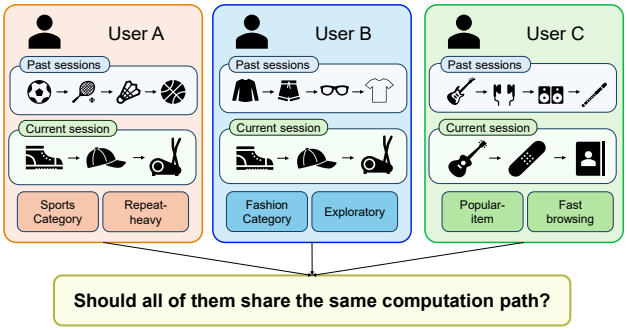

Figure 1: Different users and sessions can require different computational paths.

1 Introduction

Sequential recommendation is usually formulated as next-item prediction over an ordered interaction history, but the underlying behavior is rarely homogeneous. Even for the same user, one prefix may reflect stable preference, another may reflect an evolving within-session goal, and another may be dominated by immediate local exploration. Prior work on sequential and session-based recommendation has repeatedly shown that user behavior mixes multiple regimes rather than following a single stationary pattern [39, 40].

Strong sequential recommenders model these histories increasingly well [12, 17, 23, 26, 30, 34]. However, most of them still rely on largely shared computation across prefixes that may arise from very different behavioral contexts. This shared-path assumption is practical and often effective on average, but it also creates a modeling mismatch: heterogeneous behavior is acknowledged in the data, while computation remains mostly uniform.

Figure 1 illustrates this gap. Distinct users and sessions can differ in pace, repetition, focus, and local transition patterns, yet standard sequential models expose limited control over how such differences should change the computation path. Existing advances therefore improve representation quality, but they leave the allocation of computation mostly implicit.

Feature-aware and heterogeneity-aware sequential models partially address this issue, but mostly by enriching representations, reweighting attention, or decomposing interests inside a shared computation pipeline. In other words, prior work is not unaware of behavioral heterogeneity; it usually chooses to absorb that heterogeneity into better sequence representations. In contrast, the question we ask here is more direct: can observable behavior cues control *which computation path* should be used? Mixture-of-experts (MoE) is a natural candidate for this purpose [7, 13, 16, 33], because

it can replace one shared transformation with condition-dependent expert allocation. Yet simply inserting experts is not enough. For sequential recommendation, routing must still answer why specialization is needed in the first place, where routing should differ across temporal evidence, and what signals should drive routing in an interpretable way.

Challenge 1: behavioral diversity versus shared computation. Real prefixes arise from different behavioral regimes, but most sequential recommenders still process them with largely shared computation. This leaves the model with limited ability to specialize computation for persistent preference, evolving session intent, or immediate local transitions. At this point, the first question is simply why routing should exist at all in sequential recommendation. Our answer is that if routing is to be justified beyond generic extra capacity, it should be controlled by portable cues that can be derived from ordinary logs rather than from heavy dataset-specific side branches.

Challenge 2: temporal-scope mismatch. Once routing is introduced, another issue appears immediately: behavioral evidence arrives at different horizons. Long-range history reflects relatively stable tendency, within-session context reflects evolving intent, and short-range context reflects immediate transition pressure. A single routing view is therefore too coarse: it can blur signals that should influence expert allocation at different granularities. The second question is therefore where routing should differ. Our answer is that routing should be aligned explicitly with macro, mid, and micro temporal scopes rather than imposed as one uniform control signal.

Challenge 3: opaque hidden-state routing. Even if multi-stage routing is available, routing is often driven primarily by hidden representations. This is effective, but it makes expert allocation difficult to inspect because the same latent space must both encode sequence content and decide computation paths. The third question is what should drive routing. Our answer is that if routing is meant to reflect behavior, it should be grounded in signals that can be traced back to observable interaction patterns rather than hidden-state geometry alone.

Taken together, these challenges raise three concrete design questions: why routing should exist at all, where it should differ across temporal evidence, and what should drive it in an interpretable way.

We address these challenges with **RouteRec**, a behavior-guided MoE framework for sequential recommendation. RouteRec is organized around three design axes that map directly to the challenges above. *Portable Behavioral Cues* summarize ordinary logs into compact routing signals, providing a lightweight interface for behavior-aware control. Such cues include, for example, inter-event tempo, category concentration and switching, recurrence patterns, and popularity exposure. *Coarse-to-Fine Multi-Stage Routing* applies those cues across macro, mid, and micro scopes so that expert allocation follows temporal evidence granularity rather than a single global bias. *Behavior-Guided Expert Allocation* decouples routing signals from hidden representations, so experts transform sequence states while routing remains tied to inspectable behavior cues.

This framing deliberately treats behavioral information as a control interface for conditional computation rather than as another feature-fusion pathway. As a result, RouteRec preserves a strong sequence backbone while exposing a lightweight mechanism for behavior-grounded specialization. The rest of the paper follows this structure: Related Work clarifies where prior studies stop short of these goals, Method formalizes the three design axes, and Experiments test whether compact cues, multi-stage routing, and decoupled expert allocation each matter in practice.

The main contributions of this work are as follows:

- We identify three front-end challenges for conditional computation in sequential recommendation: the mismatch between behavioral diversity and shared computation, temporal-scope mismatch in routing, and the opacity of hidden-state-only expert allocation.
- We propose RouteRec, which combines portable behavioral cues, coarse-to-fine multi-stage routing, and behavior-guided expert allocation to align expert usage with observable behavior.
- We design RouteRec to operate from ordinary interaction logs, making the routing interface lightweight, transferable, and suitable for diagnostic analysis across datasets.
- We show that RouteRec improves recommendation quality over strong baselines on several datasets and yields more behavior-consistent routing patterns in ablation and diagnostic studies.

2 Related Work

2.1 Shared-Path Sequential Modeling and Conditional Computation

Sequential recommendation has progressed from recurrent and convolutional models to graph and Transformer architectures [12, 17, 35, 37, 41]. Recent methods further improve performance with time-aware attention, contrastive learning, and structural inductive bias [6, 23, 26, 30, 34, 38, 42]. These methods substantially strengthen sequence representation quality, but they generally preserve one shared computation pathway for behaviorally different prefixes.

Conditional computation and MoE provide a natural response to this limitation. Classical MoE and later sparse routing work show that different inputs need not share the same transformation path [7, 13, 16, 33]. Recommendation systems have also adopted expert decomposition to improve specialization and capacity [2, 24, 28, 36, 46]. However, existing sequential recommendation literature still gives limited attention to the specific question we focus on here: how to let observable behavioral diversity determine computation allocation. RouteRec addresses this gap by building a portable behavior cue interface for routing rather than keeping all heterogeneity inside a shared-path encoder.

2.2 Multi-Scope Behavioral Signals in Sequential Recommendation

Many studies explicitly acknowledge that user state is heterogeneous or non-stationary, including short-term intent modeling, multi-interest extraction, and long-short preference decomposition [5, 21, 22, 25, 27, 31, 45]. Feature-aware sequential recommenders also incorporate timestamps, side information, and auxiliary feature channels to enrich sequence encoding [1, 6, 23, 43, 47]. Together, these lines of work show that sequential behavior is expressed across multiple temporal scopes and can benefit from richer behavioral evidence.

At the same time, most prior approaches use these signals to refine hidden representations, reweight attention, or fuse side information. That is, they acknowledge multi-scope behavior, but they do not usually turn scope differences into an explicit routing design problem. RouteRec builds on this literature by asking not only how to represent multiple scopes, but also how to assign computation differently across them. Its coarse-to-fine multi-stage routing therefore treats macro, mid, and micro evidence as distinct control views rather than as one pooled feature stream.

2.3 Expert Routing Semantics and Inspectability

MoE-based recommendation models show that expert specialization can improve flexibility, but routing semantics often remain unclear. In many architectures, routing is driven by hidden representations, task identities, or learned latent facets [2, 24, 28, 36, 46, 48, 49]. This can be effective, yet it leaves a limited explanation of why a particular expert was chosen and whether that choice corresponds to an observable behavioral regime.

For our setting, this interpretability issue matters because RouteRec is not only trying to add capacity; it is trying to specialize computation according to behavior. Prior expert models mainly specialize through latent, task-level, or facet-level signals, whereas RouteRec focuses on observable cues tied to temporal behavioral regimes. This difference matters for the later experiments because it changes not only how routing is computed, but also what kinds of diagnostic questions the model can answer. We therefore separate the role of representations and routing: hidden states remain the content channel transformed by experts, while routing is controlled by portable behavioral cues derived from standard logs.

3 Method

3.1 RouteRec Overview

Figure 2 summarizes RouteRec. For each user, we observe interactions ordered by time and train the model for next-item prediction. In one training instance, the observed prefix is denoted by $x_{1:t}$ and the target by x_{t+1}. Each interaction contains at least an item identity and timestamp, and may additionally include coarse category information depending on the dataset. RouteRec retains a strong sequential backbone for encoding the prefix, but replaces shared feed-forward computation with stage-wise MoE routing.

We first describe RouteRec as a design framework and refer to the fixed paper instantiation only when concrete implementation choices matter. The model uses two signal channels. The representation channel maps the observed prefix to hidden states, which are the content representations transformed by experts. The behavior channel summarizes the same prefix into compact cues derived from ordinary logs. These cues are deliberately lightweight: they are designed as a control interface for routing rather than as another deep side-information branch. In the main text we discuss the selected paper instantiation only when that choice matters for the empirical track, while broader variants and implementation alternatives are deferred to the appendix ablations. Appendix A collects the corresponding specification details for the selected paper track.

This design directly follows the challenge structure introduced earlier. Challenge 1 motivates portable behavior cues derived from widely available logs. Challenge 2 motivates stage-wise routing over macro, mid, and micro temporal scopes. Challenge 3 motivates decoupling behavior-grounded routing from the expert transformation pathway so that expert allocation can be analyzed in terms of observable behavior. The remainder of this section formalizes how these three ideas are instantiated in one RouteRec architecture.

Let $\mathcal{T} = \{\text{macro}, \text{mid}, \text{micro}\}$ be the stage set. Let $\mathbf{h}_t^{\text{in},(s)}$ denote the hidden state entering stage s. At stage $s \in \mathcal{T}$, routing uses a stage-specific behavior feature $\mathbf{f}_t^{(s)}$, while experts consume the current stage input state:

$$\boldsymbol{\pi}_t^{(s)} = \text{softmax}\Big(\text{Router}_s(\mathbf{f}_t^{(s)})/\tau_s\Big), \tag{1}$$

$$\tilde{\mathbf{h}}_t^{(s)} = \sum_{e=1}^{E} \pi_{t,e}^{(s)} \, \text{Expert}_e^{(s)}(\mathbf{h}_t^{\text{in},(s)}). \tag{2}$$

Eq. (2) is the common stage interface. In the serial RouteRec architecture, each stage consumes the hidden state produced by the previous attention or routing block. Thus, the framework keeps one shared MoE interface, while the paper instantiation specifies how these stage-local states are connected in sequence.

3.2 Portable Behavioral Cues

RouteRec is designed for realistic data availability. It uses signals that are commonly present in interaction logs: item identity, timestamp, optional coarse item-group labels, and corpus-level popularity statistics derived from the training data. We intentionally avoid heavy side-information encoders in the routing branch. Instead, raw signals are converted into compact scalar summaries that are easy to compute, stable across datasets, and interpretable at analysis time.

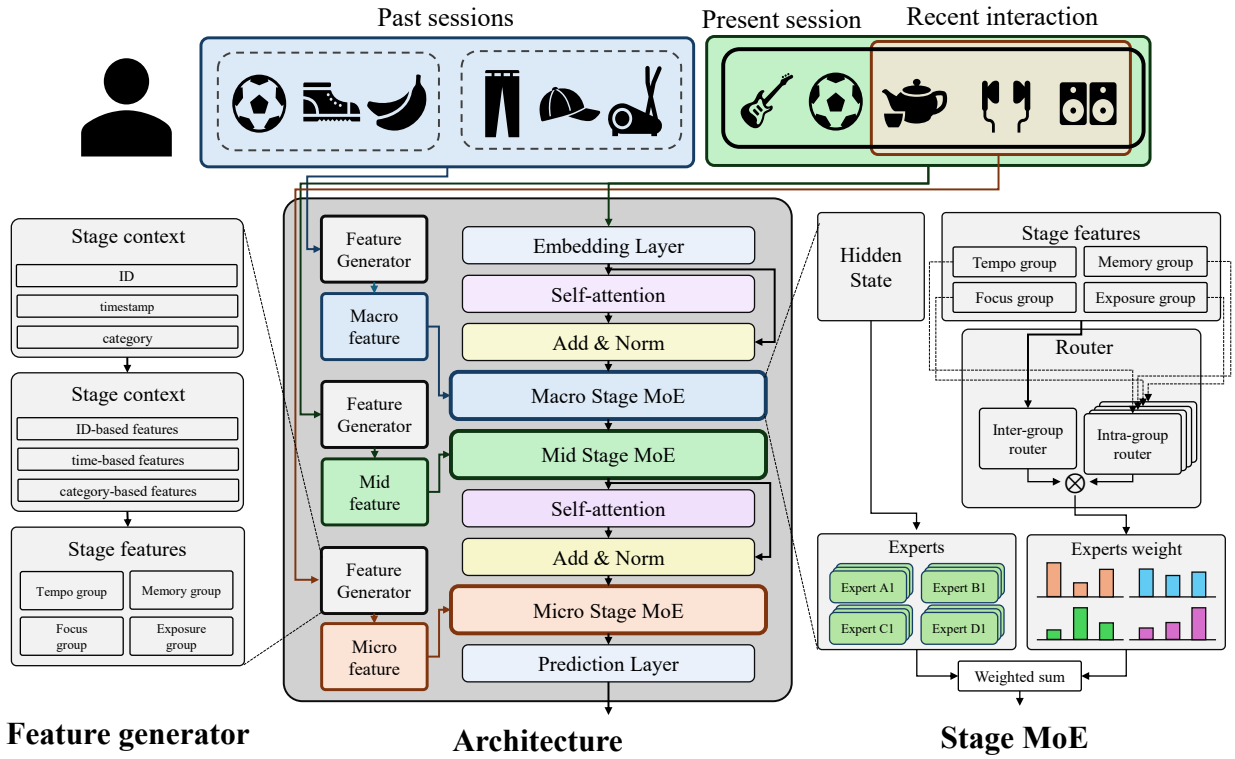

Figure 2: Overall RouteRec framework. (a) RouteRec keeps a strong sequential backbone and replaces shared feed-forward blocks with behavior-guided MoE stages. (b) All stages use the same scalar-group conditional routing composition: a group-level distribution and a conditional intra-group distribution are multiplied to form expert probabilities. (c) Macro, mid, and micro differ by temporal evidence scope and routing granularity, not by router formula.

For session s_m and target position t, the stage windows are

$$\mathcal{R}_t^{\text{macro}} = [s_{\max(1,m-H)}, \ldots, s_{m-1}], \tag{3}$$

$$\mathcal{R}_t^{\text{mid}} = [x_1, \ldots, x_{t-1}], \tag{4}$$

$$\mathcal{R}_t^{\text{micro}} = [x_{\max(1,t-w)}, \ldots, x_{t-1}], \tag{5}$$

where $H = 5$ and $w = 5$ in our main setting. These are the default paper windows used throughout the main experiments, and their role is revisited in the structural ablations in Appendix C. A shared summary operator $\mathcal{A}(\cdot)$ maps each window to a low-dimensional feature vector:

$$\mathbf{f}_t^{(s)} = \mathcal{A}(\mathcal{R}_t^{(s)}), \qquad s \in \mathcal{T}.$$

The summary dimensions are grouped into four semantic families: *Tempo*, *Focus*, *Memory*, and *Exposure*. Tempo captures coverage, pace, and interval dynamics. Focus captures concentration and switching behavior over the coarsest item-group proxy available in the dataset, such as product category, venue category, artist grouping, or genre. Memory captures repetition, overlap, and recurrence structure. Exposure captures the level, spread, and drift of interacted popularity. In the fixed paper instantiation, this family template yields a compact 16-dimensional cue bank per stage. When such grouping fields are unavailable, the corresponding focus cues are handled by the fixed missing-signal rule documented in Appendix A. Representative examples are listed in Table 4.

Importantly, the feature template is shared across stages, but each stage reads it through a different temporal lens. Macro features reflect persistent tendency and broad preference shape. Mid features track evolving intent inside the current session. Micro features emphasize immediate cues around the prediction target. This shared-template, stage-specific-view design keeps the feature branch compact while preserving temporal specificity.

This feature design is intentionally more structured than generic feature fusion. The goal is not to expose as many handcrafted statistics as possible, but to construct a small cue bank whose dimensions remain semantically legible and transportable across datasets. We therefore use stage-specific summaries with a consistent family template, rather than dataset-specific deep side branches. This family structure is treated as a semantic prior rather than as an immutable ontology, and Section 4.4 tests whether that organization matters beyond using an unstructured bag of scalars. The appendix records representative cue definitions and the fixed

paper setting, while more exhaustive column-level formulas can remain outside the main text because they are engineering details rather than the conceptual core of RouteRec.

3.3 Coarse-to-Fine Multi-Stage Routing

Although the router formula is shared, stages are not redundant. They differ in temporal evidence, routing granularity, and placement inside the backbone. In the fixed paper instantiation, the serial layout is

$$\text{self-attention} \rightarrow \text{macro routing block} \rightarrow \text{mid routing block} \rightarrow \text{self-attention} \rightarrow \text{micro routing block}.$$

Let $\mathbf{h}_t^{(0)}$ denote the hidden state after the first attention block, $\mathbf{h}_t^{(1)}$ after macro routing, $\mathbf{h}_t^{(2)}$ after mid routing, and $\mathbf{h}_t^{(3)}$ after the second attention block. Macro, mid, and micro therefore consume $\mathbf{h}_t^{(0)}$, $\mathbf{h}_t^{(1)}$, and $\mathbf{h}_t^{(3)}$, respectively. The first attention layer builds a contextualized prefix representation before any routing decision is made. Macro and mid then operate consecutively because both provide slower sequence-level control over closely related contextual states: macro reflects broader recent-history tendency, whereas mid refines the evolving intent of the current session. In our architecture sweep, inserting another attention layer between them increased depth without yielding a comparably distinct temporal view; the issue is not depth for its own sake, but whether the extra update creates a meaningfully different temporal control view. By contrast, the second attention layer is placed immediately before micro so that the final local router sees a refreshed token interaction state before prediction. Appendix C reports the corresponding stage-order and layout ablations for this selected architectural choice.

Macro and mid (sequence-wise routing). Macro and mid compute one routing distribution per sequence from their respective windows, then broadcast it over valid positions inside the sequence. Macro uses broader cross-session history; mid uses within-session prefix evidence. This setting emphasizes stable allocation for slower-changing context, where token-wise routing tended to add variance without introducing an equally meaningful control resolution in our ablations.

Micro (position-wise routing). Micro computes routing per token near the prediction target using short-horizon context. This enables fast adaptation to immediate transitions without changing the router form itself.

Formally, let $\bar{\mathbf{f}}_i^{(s)}$ denote the sequence-level cue summary used for sequence-wise routing at stage s. Then macro and mid use

$$\bar{\boldsymbol{\pi}}_i^{(s)} = \mathrm{softmax}\Big(\mathrm{Router}_s(\bar{\mathbf{f}}_i^{(s)})/\tau_s\Big), \qquad s \in \{\text{macro}, \text{mid}\},$$

$$\boldsymbol{\pi}_{i,t}^{(s)} = \bar{\boldsymbol{\pi}}_i^{(s)}.$$

Micro instead applies the same router form directly to position-specific cues,

$$\boldsymbol{\pi}_{i,t}^{(\text{micro})} = \mathrm{softmax}\Big(\mathrm{Router}_{\text{micro}}(\mathbf{f}_{i,t}^{(\text{micro})})/\tau_{\text{micro}}\Big).$$

Hence, RouteRec's coarse-to-fine behavior comes from stage evidence scope and update frequency, while the underlying router composition remains constant. The stages are therefore not repeated copies of the same router; they correspond to different temporal horizons at which behavioral evidence becomes informative for computation allocation.

3.4 Behavior-Guided Expert Allocation

The final RouteRec architecture uses one shared scalar-group conditional composition at every stage. Experts are partitioned into G groups with C experts per group, so $E = GC$. For stage s and position t, the router produces

$$p_{t,g}^{(s)} = \mathrm{softmax}_g\Big(z_{t,g}^{(s)}\Big), \tag{6}$$

$$r_{t,g,c}^{(s)} = \mathrm{softmax}_c\Big(u_{t,g,c}^{(s)}\Big), \tag{7}$$

where $p_{t,g}^{(s)}$ is a group-level routing distribution and $r_{t,g,c}^{(s)}$ is a conditional within-group routing distribution. The final expert probability is their product:

$$\pi_{t,g,c}^{(s)} = p_{t,g}^{(s)}\, r_{t,g,c}^{(s)}, \tag{8}$$

followed by flattening (g, c) to expert index e. Eq. (8) is the same for macro, mid, and micro.

In the paper instantiation, the group structure is organized around the four cue families, while capacity is expanded by multiple experts within each group. This alignment operates only at the router-prior level: it defines a semantic prior over expert organization rather than a hard hand assignment of individual cues or cue dimensions to individual experts. The experts themselves remain fully learned.

In the main configuration, all three stages use the same feature-driven group-conditional router. A group-level scalar router first determines which semantic control view is most relevant. A conditional intra-group router then distributes probability mass among the experts inside that group. In plain terms, RouteRec first selects the family-level control view and then allocates capacity within that view. This is the core reason we describe the method as behavior-guided allocation: the router is controlled by the cue bank, while hidden states remain the objects transformed by the experts. We choose dense soft mixing in this paper because it makes stage-wise specialization easier to compare and diagnose across datasets [48, 49]; sparse top-k routing remains an orthogonal systems choice outside the present scope.

This unified composition is important for two reasons. First, it gives all stages a shared structural prior, which reduces stage-specific router engineering and simplifies debugging. Second, it keeps diagnostics comparable across stages because differences in routing behavior can be attributed to evidence scope rather than formula changes.

In RouteRec, routing is driven by behavior cues, while hidden states remain dedicated to expert transformation in Eq. (2). This preserves the routing-representation decoupling principle that underlies our approach.

3.5 Training Objective

RouteRec is trained with a standard next-item objective and two router-side regularizers in the main paper configuration:

route consistency and z-loss. The first regularizer shapes routing semantics, and the second stabilizes routing numerics.

Let $\mathbf{z}$ be final logits and y be the target item. The main loss is

$$\mathcal{L}_{\mathrm{CE}} = -\log \frac{\exp z_y}{\sum_j \exp z_j}.$$

For route consistency, we first compute a sample-level routing distribution at each stage by averaging token-level routing weights:

$$\bar{\boldsymbol{\pi}}_i^{\,s} = \frac{1}{L_i} \sum_{t=1}^{L_i} \boldsymbol{\pi}_{i,t}^{\,s}$$

where L_i is the valid sequence length. We also form a stage-level cue summary

$$\bar{\mathbf{f}}_i^{\,s} = \frac{1}{L_i} \sum_{t=1}^{L_i} \mathbf{f}_{i,t}^{\,s}.$$

For each stage s, we construct a pair set $\mathcal{P}_s$ from the k_{nn} nearest non-self samples in the mini-batch according to cosine similarity between these stage-level cue summaries, and minimize

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{|\mathcal{P}_s|} \sum_{(i,j) \in \mathcal{P}_s} \mathrm{JSD}\big(\bar{\boldsymbol{\pi}}_i^{\,s}, \bar{\boldsymbol{\pi}}_j^{\,s}\big)\,,$$

which encourages neighboring behavioral states to produce compatible routing distributions. In other words, semantically similar inputs should not induce arbitrarily different routing behavior. Appendix A reports the neighbor setting used in the paper instantiation. For macro and mid, the sample-level average reduces to the broadcast sequence-wise route; we keep the averaging notation for a unified treatment across stages.

To stabilize gating numerics, we adopt z-loss on the pre-softmax expert logits $a_{i,t,e}^{\,s}$ [49]:

$$\mathcal{L}_z = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{N_s} \sum_{(i,t) \in \mathcal{V}_s} \left(\log \sum_{e=1}^{E} \exp a_{i,t,e}^{\,s} \right)^2,$$

where $\mathcal{V}_s$ is the set of valid positions and $N_s = |\mathcal{V}_s|$. This term discourages excessive logit scale and helps prevent unstable, overly sharp routing. The final objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{cons}}\, \mathcal{L}_{\mathrm{cons}} + \lambda_z\, \mathcal{L}_z,$$

where λ_{cons} and λ_z are trade-off coefficients. This objective jointly optimizes prediction quality, semantic consistency of routing, and numerical stability of expert selection. Unlike classical MoE setups that explicitly enforce uniform expert traffic, the selected main configuration does not include a load-balancing term in the final paper runs. Our goal here is not uniform usage for its own sake, but stable and behavior-aligned specialization. Appendix E reports objective variants including optional balance regularization.

4 Experiments

4.1 Experimental Setup

We evaluate RouteRec on six public datasets. These datasets cover e-commerce, location recommendation, short-video consumption, music listening, movie preference, and anonymous clickstream behavior. The six corpora are Beauty, Foursquare, KuaiRec, LastFM, ML-1M, and Retail Rocket. Appendix B summarizes their provenance and processed statistics.

Across datasets, we follow a sessionized temporal evaluation protocol assembled from established practices in session-based and session-aware recommendation rather than claiming a single canonical raw-sequence benchmark [14, 20, 31]. Interactions are first ordered chronologically, converted to sessions with inactivity-based heuristics, and then filtered to remove sessions shorter than 5 and items with fewer than 3 interactions, which is consistent with common session-based benchmark stabilization pipelines [22, 41]. We use a 30-minute threshold for Foursquare, KuaiRec, LastFM, ML-1M, and Retail Rocket, which matches common practice in dense timestamped logs and later session-aware studies [18, 31], whereas Beauty uses a 14-day threshold because review timestamps are much sparser than click, play, or check-in logs. This follows the broader observation that sparse or item-poor domains may require task-dependent session definitions rather than a universal short timeout [3]. Beauty is constructed from Amazon Beauty reviews and additionally excludes interactions with rating below 3.

For transparency, the reported KuaiRec and LastFM results are obtained from processed sampled subsets rather than the full raw corpora: specifically, we use a 20% KuaiRec subset and a 3% LastFM subset, and we mark these datasets explicitly in the result tables. Sessions are then sorted by start time and split chronologically into 70%/15%/15% train/validation/test partitions. This is a deliberate methodological choice: sessions are the unit of evaluation because the paper studies session-aware behavioral routing, the split preserves temporal order at the session level, and the held-out construction is designed to keep the benchmark auditable rather than to mimic a raw-sequence leaderboard directly. Because recent work has shown that split construction can materially affect offline conclusions and may induce leakage when chronology is ignored, we keep the temporal protocol explicit and reserve the full preprocessing details for Appendix B [10, 15].

Our main comparison uses the seen-target evaluation view. This choice is deliberate: ID-based next-item recommenders cannot rank items absent from the training vocabulary, and session-aware benchmarks commonly treat unseen targets as a supplementary rather than primary evaluation view [20, 22, 41]. Concretely, held-out cleanup removes unseen non-target rows from the primary files, while unseen targets are retained only for supplementary accounting rather than the main comparison. Appendix B gives the exact held-out cleanup procedure, the processed dataset statistics, the tuning protocol, and the full $k \in \{5, 10, 20\}$ tables.

The baselines include GRU4Rec, SASRec, TiSASRec, DuoRec, FEARec, BSARec, FAME, DIF-SR, and FDSA [6, 12, 17,

Table 1: Seen-target test results under the sessionized temporal protocol. Datasets marked with † are sampled processed subsets (KuaiRec 20%, LastFM 3%). For each method, we report the test metrics of the run achieving the highest mean validation score over the nine reported metrics. Avg. Rank is computed over all rows in this table, where lower is better. The RouteRec column is shaded for readability.

Dataset	Metric	SASRec	GRU4Rec	TiSASRec	FEARec	DuoRec	BSARec	FAME	DIF-SR	FDSA	RouteRec
Beauty	HR@10	0.1347	0.0716	0.1318	0.1261	0.1146	0.0516	0.0315	0.1060	0.1347	0.1616
	NDCG@10	0.0748	0.0340	0.0878	0.0696	0.0680	0.0316	0.0211	0.0685	0.0909	0.0982
	MRR@20	0.0588	0.0247	0.0770	0.0544	0.0569	0.0260	0.0190	0.0592	0.0805	0.0822
Retail Rocket	HR@10	0.5605	0.5211	0.5435	0.5441	0.5377	0.4703	0.4697	0.4365	0.5133	0.5612
	NDCG@10	0.4099	0.3858	0.3963	0.3935	0.3864	0.3963	0.3934	0.3900	0.3651	0.4150
	MRR@20	0.3673	0.3479	0.3547	0.3515	0.3441	0.3743	0.3710	0.3744	0.3239	0.3737
Foursquare	HR@10	0.3233	0.2130	0.3157	0.3042	0.3197	0.2462	0.2546	0.3034	0.2998	0.3225
	NDCG@10	0.2014	0.1455	0.2016	0.1925	0.1955	0.1639	0.1646	0.1984	0.2001	0.2041
	MRR@20	0.1679	0.1278	0.1700	0.1619	0.1607	0.1410	0.1401	0.1690	0.1723	0.1712
ML-1M	HR@10	0.1478	0.1413	0.1538	0.1487	0.1520	0.1589	0.1557	0.1501	0.1594	0.1561
	NDCG@10	0.0770	0.0742	0.0844	0.0724	0.0811	0.0854	0.0853	0.0764	0.0835	0.0771
	MRR@20	0.0608	0.0587	0.0690	0.0550	0.0649	0.0678	0.0687	0.0595	0.0658	0.0585
LastFM†	HR@10	0.3994	0.3090	0.3780	0.3765	0.3831	0.3453	0.3312	0.3617	0.3657	0.3794
	NDCG@10	0.3267	0.2673	0.3196	0.3167	0.3083	0.3024	0.2900	0.3149	0.3171	0.3253
	MRR@20	0.3054	0.2561	0.3027	0.2989	0.2863	0.2902	0.2784	0.3020	0.3033	0.3099
KuaiRec†	HR@10	0.1055	0.0945	0.1011	0.1005	0.0995	0.0918	0.1011	0.0918	0.0967	0.1060
	NDCG@10	0.0943	0.0790	0.0916	0.0919	0.0919	0.0890	0.0915	0.0871	0.0865	0.0965
	MRR@20	0.0922	0.0753	0.0894	0.0903	0.0905	0.0882	0.0899	0.0863	0.0843	0.0948
Avg. Rank↓		3.42	9.06	3.72	5.92	5.69	6.56	6.86	6.42	5.14	2.22

23, 24, 30, 34, 43, 47]. All methods are trained within the same optimization family, with matched early-stopping rules and comparable capacity ranges. In the comparison tables, each baseline is represented by the run attaining the highest mean validation score over the nine reported metrics, and we report the corresponding test metrics. RouteRec is reported from the best validation-selected configuration considered in this study under the same criterion.

Given that ranking conclusions in sessionized sequential recommendation can change with both protocol construction and optimization quality [10, 15, 19, 20], we treat tuning as part of the experimental protocol rather than as a secondary implementation detail. Concretely, each model is tuned within a predefined model-specific discrete grid covering standard optimization hyperparameters and a small number of architecture-specific controls. The admissible values are fixed in advance; when adaptive search is used for efficiency, it is applied only over these predeclared candidate sets, following the same validation files, early-stopping rule, and selection criterion for all methods [8]. The intended comparison is therefore transparent and controlled: adaptive search is used only as an efficient implementation over fixed candidate sets, not as open-ended retuning. Appendix B summarizes the bounded candidate grids and trial budgets used to audit this procedure.

All methods are trained and evaluated on the same processed train/validation/test files. RouteRec does not access different held-out sessions or external metadata at evaluation time; its routing cues are computed from the same observed interaction histories that define the benchmark instances. The intended comparison is therefore a sessionized, behavior-aware next-item setting under a shared temporal protocol, not a claim about direct comparability to canonical raw-sequence leaderboards.

The main experimental section is organized around five questions: **Q1)** whether RouteRec improves ranking quality under the shared protocol, **Q2)** what should control routing, **Q3)** why the three-part design is effective, **Q4)** whether lightweight cues are sufficient in practice, and **Q5)** whether the learned routing patterns align with behavioral regimes. This organization keeps the main text focused on the core empirical claims, while the appendix provides broader controls, definitions, and supplementary analyses.

4.2 Q1. Does RouteRec Improve Ranking Quality?

Table 1 presents the main seen-target comparison in the text using HR@10, NDCG@10, and MRR@20, together with an Avg. Rank summary across all reported rows. This presentation keeps the main paper focused on a compact set of representative cutoffs, while Appendix B reports the full $k \in \{5, 10, 20\}$ grid under the same validation-based selection rule. RouteRec exhibits its clearest gains on Beauty and

(a) Avg. ranking quality across routing controls

(b) Route consistency over feature-similarity buckets

Figure 3: Routing-control comparison. (a) Average MRR@20 across shared FFN, hidden-only, mixed, and behavior-guided routing. (b) Routing similarity over feature-similarity buckets.

the sampled KuaiRec subset, remains competitive on Retail Rocket, and stays close to the strongest shared-backbone baselines on Foursquare, LastFM, and ML-1M.

Q1 establishes the aggregate competitiveness of the full model under the shared protocol and selection rule. The main issue is not whether every dataset is won uniformly, but whether the overall performance pattern is strong enough to motivate the subsequent design-oriented analyses.

4.3 Q2. What Should Control Routing?

Figure 3 compares four routing controls: a shared FFN, hidden-only routing, mixed hidden+behavior routing, and the final behavior-guided router. Hidden-only routing can improve over a shared FFN, but the gain is smaller and less stable than behavior-guided routing. Re-injecting hidden states into the router also fails to produce a consistent advantage over the behavior-only design.

Q2 isolates the source of the routing decision. In particular, it examines whether the gains can be recovered by a stronger hidden-state gate, or whether explicit behavioral control provides a qualitatively different advantage. The consistency panel offers a complementary view: if routing is grounded in observable cues rather than hidden-state geometry alone, behaviorally similar cases should induce more compatible expert patterns. Because RouteRec also includes a routing-side regularizer, Appendix E reports the corresponding objective variants to separate routing-signal effects from regularizer effects. This analysis directly addresses Challenge 3 and supports the behavior-guided expert-allocation axis introduced in Section 3.

4.4 Q3. Why Is RouteRec's Three-Part Design Effective?

Figure 4 tests whether RouteRec's gains arise from the integrated three-part design itself, namely semantic cue organization, staged temporal allocation, and the final router composition, rather than from indiscriminate expert insertion. Panel (a) tests semantic cue design variants such as reduced, shuffled, flat, and random-group settings rather than merely the presence of extra scalar inputs. Panel (b) tests stage-semantics variants including scope-swap, identical-scope, and layout controls rather than merely the presence of multiple MoE blocks. Panel (c) tests router-composition variants including stage placement, ordering, and router alternatives rather than just one arbitrary implementation among several similar choices.

Taken together, these ablations examine whether the gains arise from one isolated component or from the interaction of semantic cue design, stage semantics, and router composition. Q3 therefore addresses the internal coherence of the proposed design, whereas Q4 addresses its robustness under limited cue availability. More broadly, it tests whether RouteRec benefits from the interaction of cue semantics, temporal scope decomposition, and behavior-guided allocation rather than from one isolated trick. This analysis directly addresses Challenge 2 and supports the coarse-to-fine multi-stage routing axis.

4.5 Q4. Are Lightweight Cues Sufficient in Practice?

Figure 5 evaluates whether RouteRec depends on rich metadata or whether ordinary interaction logs already provide enough routing evidence. Unlike Q3, which asks whether the semantic cue design is meaningful, Q4 asks a portability question: how much observable signal must remain available before behavior-guided routing stops being useful. The left panel therefore focuses on availability-oriented reductions such as removing category cues, removing time cues, or retaining only sequence-derived summaries, and the right panel normalizes each variant by the gain of the full-cue model.

This analysis evaluates the portability claim in a conservative setting. While RouteRec may benefit from richer evidence when it is available, the main question is whether a practically deployable subset of cues can preserve a meaningful fraction of the full model's benefit. This is the main test of the claim that ordinary logs already expose enough routing evidence for behavior-guided specialization to remain useful. This analysis directly addresses Challenge 1 and supports the portable behavioral-cue axis.

4.6 Q5. Do Routing Patterns Align with Behavioral Regimes?

Figure 6 addresses the most direct semantic question raised by the method: if RouteRec is designed to react to behavioral regimes, then its gains and routing sharpness should appear where those regimes differ most. We therefore summarize results over repeat-heavy, high-tempo, narrow-focus, and exploration-heavy slices, and compare both ranking quality and routing concentration within those regimes.

This analysis connects the empirical evidence back to the motivation in Figure 1. To avoid circularity, the slice definitions are specified explicitly in Appendix H using simple post hoc statistics derived from raw interaction histories, such as repeat ratio, average inter-event gap, concentration entropy, and switching or novelty rate, rather than optimized labels or

(a) Semantic cue design variants (b) Stage-semantics variants (c) Router composition variants

Figure 4: Three-part design ablations. (a) Semantic cue design variants. (b) Stage-semantics variants. (c) Router composition variants.

(a) Avg. MRR@20 under reduced cue availability (b) Relative retention of the full-cue gain

Figure 5: Lightweight-cue analysis. (a) Average MRR@20 under reduced cue availability. (b) Gain retention relative to the full cue set.

(a) Slice-wise MRR@20 across behavioral regimes (b) Relative gain and routing concentration by slice

Figure 6: Behavior-regime analysis. (a) Slice-wise ranking quality across repeat-heavy, fast-tempo, focused, and exploration-heavy regimes. (b) Relative RouteRec gain and routing concentration across the same slices.

supervision targets. The goal is not to show that RouteRec merely recovers its own features, but that externally defined behavioral regimes exhibit systematically different gain and routing-sharpness patterns.

Detailed full-cutoff results and extended analyses, including broader structural ablations, protocol-sensitivity checks, objective variants, efficiency summaries, routing diagnostics, qualitative cases, and transfer studies, are reported in Appendices A–I.

5 Conclusion

RouteRec starts from a simple hypothesis: when behavioral evidence changes, the computation path should change as well. The experiments support this view from three main angles. First, RouteRec improves next-item ranking most clearly on behaviorally dynamic datasets and remains competitive on several others, although the advantage narrows in more preference-stable settings such as ML-1M. Second, the gain is tied to the routing signal and the staged design rather than to indiscriminate expert insertion. Third, lightweight cues derived from ordinary interaction logs can preserve a meaningful share of the benefit, which keeps the method practical even when side information is limited.

Taken together, these results suggest that compact behavioral cues provide a useful control interface for conditional computation in recommendation, rather than merely another feature-fusion pathway. The main contribution is therefore the combination of three design axes, portable behavioral cues, coarse-to-fine multi-stage routing, and behavior-guided expert allocation, into one coherent behavior-conditioned computation design. Future work can build on this interface with broader transfer regimes, richer but still robust cue construction, stronger sparse-routing variants, and tighter deployment-aware constraints such as latency, continual adaptation, and router calibration.

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Table 2: Fixed main configuration used in the paper.

Aspect	Main paper configuration
Backbone layout	self-attention → macro routing block → mid routing block → self-attention → micro routing block
Stage execution	Serial
Routing granularity	Macro and mid use sequence-wise routing; micro uses position-wise routing
Router composition	Shared scalar-group conditional router at all three stages
Expert mixing	Dense softmax mixture over the full expert set
Macro history window	$H = 5$ recent sessions
Core auxiliary terms	Route consistency + z-loss; load balancing disabled in the selected paper runs

A Method Details and Design Rationale

This appendix records the fixed paper instantiation and the representative cue bank used by RouteRec. Whereas the main text emphasizes the conceptual design axes, this section documents the implementation choices that are relevant for reproducibility and for interpreting the later ablations.

A.1 Main Paper Configuration

The placement of the two attention layers reflects the intended temporal roles of the stages. The first attention layer contextualizes the raw prefix before any routing is applied. Macro and mid then operate consecutively because both provide slower sequence-level control over the same contextualized sequence state: macro captures broader recent-history tendency, while mid refines within-session evolution. A second attention layer is inserted only before micro so that the final local router receives a refreshed token interaction state immediately before prediction. This layout was the most stable high-performing architecture in our architecture sweep, and Appendix C compares it against denser and differently ordered alternatives.

This table serves as a reproducibility summary rather than as a search-space description. Architectural choices are fixed across the paper track, whereas capacity and optimization values are selected per dataset from a narrow candidate set and are therefore reported as compact ranges only when no single global value is shared across all reported runs.

A.2 Implementation Specification

A.3 Representative Behavioral Cue Families

This cue bank is intentionally compact. The feature families are hand-designed in the sense that they encode behavioral hypotheses, but they are not intended to become an unbounded feature-engineering exercise. Instead, the design goal is to construct a small set of summaries that can serve as a stable routing interface across datasets. In practice, the macro bank is instantiated from the recent-session window used by the selected paper instantiation, while mid and micro are computed from the current session prefix at different temporal scopes. The main text therefore reports the family structure and representative examples, while this appendix provides the implementation-oriented view.

Table 3: Implementation-level settings used in the selected paper track.

Item	Reported setting
Semantic groups	$G = 4$ groups aligned with Tempo, Focus, Memory, and Exposure
Experts per group	$C \in \{2, 3, 4\}$ in the paper track; total experts $E = 4C$
Cue dimensionality	16 scalars per stage in the fixed paper instantiation
Router hidden width	Selected from a small candidate set in the paper track; representative range 48–128
Routing temperature	Macro 1.0; mid and micro 1.2 in the reported setting
Consistency neighbors	$k_{\mathrm{nn}} = 1$ nearest non-self sample per stage in the paper track
Missing-signal handling	Broadcast scalar fields across positions when available; zero-fill group-based focus cues when the grouping field is unavailable

B Dataset, Evaluation, and Full Results

This appendix summarizes the six datasets and makes the evaluation protocol explicit, since recent work has shown that session construction and split strategy can materially affect offline recommender results [10, 15]. The original releases are cited here for completeness [4, 9, 11, 29, 32, 44]. The dataset descriptions emphasize domain differences and signal availability because the routing branch depends on which behavioral evidence is exposed in each corpus.

B.1 Datasets and Protocol

Table 5 reports the statistics of the final processed datasets used in the experiments rather than the raw corpus sizes. In particular, KuaiRec corresponds to the 20% processed subset used in our runs, and LastFM corresponds to the 3% processed subset.

For reproducibility, the protocol is summarized as follows.

(1) Sort all interactions chronologically within each dataset.
(2) Reconstruct sessions with a dataset-specific inactivity threshold.
(3) Iteratively prune sessions with length below 5 and items with support below 3 until the constraints are simultaneously satisfied.
(4) Sort the resulting sessions by session start time.
(5) Assign the earliest 70% of sessions to train and treat the latest 30% as a held-out tail.
(6) Build validation and test from the held-out tail under chronology-preserving cleanup.
(7) Report seen-target metrics as the primary view and keep unseen-target behavior as supplementary accounting.

Table 4: Representative cue summaries used by RouteRec in the main configuration. Each stage uses four semantic families with four scalar summaries per family; the table lists representative examples rather than every implementation detail.

Family	Macro (recent-session tendency)	Mid (within-session evolution)	Micro (short-horizon pressure)
Tempo	Context-valid ratio, last inter-session gap, mean pace, pace trend	Valid-prefix ratio, mean/std interaction interval, session age	Last gap, short-window mean gap, short-window std, deviation from mid-scale pace
Focus	Theme entropy, dominant-theme mass, theme repeat rate, theme shift rate	Category entropy, category top-1 mass, category switch rate, category uniqueness	Immediate category switch, mismatch to recent category history, suffix entropy, suffix uniqueness
Memory	Repeat intensity, adjacent category overlap, adjacent item overlap, repeat trend	Item uniqueness, repeat rate, novelty rate, longest item run	Reconsumption flag, suffix reconsumption rate, suffix item uniqueness, suffix longest run
Exposure	Mean popularity, popularity spread, popularity entropy, popularity trend	Mean popularity, popularity std, popularity entropy, popularity drift	Last-item popularity, suffix popularity spread, suffix popularity entropy, deviation from mid-scale popularity

Table 5: Datasets used in the experiments. Datasets marked with † are sampled processed subsets.

Dataset	Domain	Interactions	Sessions	Items	Avg. len.
Beauty	E-com.	33,488	4,243	3,625	7.9
Foursquare	POI	145,238	25,369	30,588	5.7
KuaiRec†	Video	287,411	24,458	6,477	11.8
LastFM†	Music	470,408	25,089	52,510	18.8
ML-1M	Movie	575,281	14,539	3,533	39.6
Retail Rocket	Browse	821,243	153,092	90,211	5.4

Beauty and Retail Rocket stress sparse e-commerce revisit behavior, Foursquare and KuaiRec emphasize stronger short-horizon intent changes, and LastFM and ML-1M provide longer preference-driven histories. This spread makes it possible to examine whether behavior-guided routing is primarily beneficial in highly dynamic settings or remains competitive when preference structure is more stable.

Signal availability also differs in a way that matters for routing. Beauty exposes item, time, and product-category cues after low-rating removal. Foursquare includes item, timestamp, and venue-category structure with bursty check-in dynamics. KuaiRec offers item, time, and coarse content-side metadata in short-video sessions with strong short-term intent shifts. LastFM provides repeated listening histories with artist-side signals, while ML-1M contributes longer preference-driven sequences with genre information. Retail Rocket is the sparsest side-information setting, relying mainly on anonymous event streams with item and time signals.

We position our setup as a sessionized sequential evaluation protocol assembled from established components, not as the canonical raw-sequence protocol for sequential recommendation [14, 20, 31]. Sessions are reconstructed post hoc with inactivity heuristics, which is standard practice when logs do not provide explicit session boundaries [14]. The thresholds are adapted to the logging regime of each corpus: Foursquare, KuaiRec, LastFM, ML-1M, and Retail Rocket use a 30-minute inactivity threshold, consistent with dense-log session-aware practice [18, 31], whereas Beauty uses a 14-day threshold because review timestamps are substantially more diffuse. More generally, this threshold choice reflects the observation that sparse and item-poor domains often require longer or more domain-specific notions of a session than dense clickstream data [3]. After sessionization, we repeatedly remove sessions shorter than 5 and items with fewer than 3 interactions until no further pruning is required. This iterative filtering follows the same general rationale as common session-based benchmark preprocessing, where short sessions and very low-support items are filtered to stabilize next-item supervision and evaluation [22, 41].

Session splits are constructed at the session level after sorting by session start time. The earliest 70% of sessions are assigned to training, and the remaining 30% are treated as a held-out tail. This tail is first divided into validation and test using stratification over three factors: whether the target item already appears in the prefix, the target item's popularity bucket with respect to the training split, and the session-length bucket. For the final evaluation files, we then merge the held-out sessions, restore chronological order, and re-split them contiguously into equal validation and test halves. This chronology-preserving design is meant to reduce leakage and keep the offline protocol aligned with temporal deployment order [10, 15].

For the primary evaluation view, we report seen-target metrics only. Concretely, in the held-out files, non-target rows whose items are unseen in training are removed, and unseen targets are retained only for supplementary accounting rather than the main comparison. This matches long-standing session-based and session-aware evaluation practice for ID-based recommenders, which cannot rank targets outside the training vocabulary and therefore typically separate unseen-target behavior from the primary benchmark table [20, 22, 41]. We report HR, NDCG, and MRR at $k \in \{5, 10, 20\}$ for all methods.

For the result tables, each baseline is represented by the run with the highest mean validation score over the nine reported metrics, and RouteRec is reported from the best validation-selected configuration under the same selection rule.

All methods use the same processed train/validation/test partitions. RouteRec's additional macro, mid, and micro cues are deterministic summaries derived from those same

observed histories, so the comparison should be read as evaluating different ways of exploiting the shared sessionized interaction context rather than different access to held-out information.

B.2 Model Selection and Tuning Protocol

Because this benchmark is built from a sessionized temporal protocol rather than a canonical raw-sequence leaderboard, the model-selection procedure is stated explicitly. Prior benchmark studies have shown that both baseline quality and model ranking can change materially once methods are re-evaluated under a common framework and retuned under a shared protocol [19, 20]. Accordingly, the comparison does not rely on paper defaults alone.

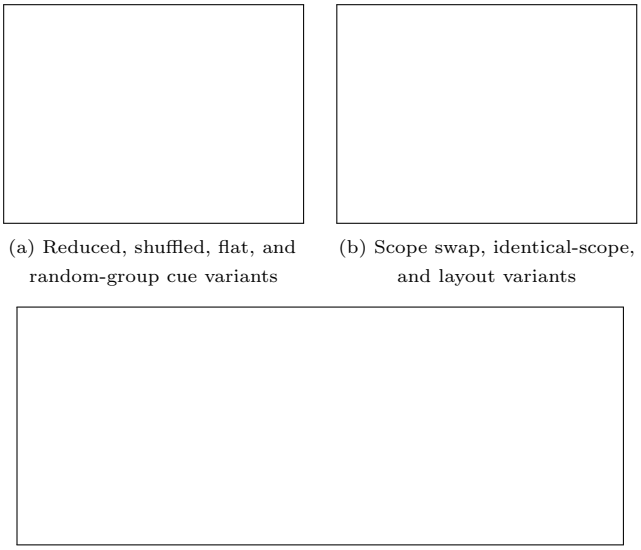
Our reporting protocol follows three simple rules. First, every model is tuned within a predefined bounded search space that is fixed before the final comparison: learning rate is sampled log-uniformly within a declared dataset-specific interval, while the remaining hyperparameters are drawn from small predeclared candidate sets. Second, these bounded spaces cover a shared core of optimization hyperparameters together with a small number of model-specific controls, so that the search is broad enough to avoid trivial under-tuning but still budgeted tightly enough to remain comparable across methods. Third, the final reported result for every model is the test performance of the configuration that maximizes the mean validation score over the nine reported metrics. When adaptive search is used for computational efficiency, it is restricted to the same predeclared bounds and candidate sets and should therefore be read as an efficient implementation of a controlled bounded search rather than as open-ended retuning [8].

The final paper track therefore uses bounded search spaces rather than unconstrained retuning. In the paper-facing comparison, baselines and RouteRec are aligned to the same shared optimization pool, and model families differ only through a small number of architecture-specific additions. For RouteRec, the selected three-stage routing layout remains fixed, but depth, hidden dropout, and routing-side controls are still tuned within bounded candidate sets. To keep the appendix concise but still reproducible, Table 6 separates the shared optimization pool from the baseline-family and RouteRec-specific additions, and lists the dataset-specific learning-rate intervals explicitly.

B.3 Full Ranking Results

Table 7 restores the full cutoff grid summarized more compactly in Table 1. This presentation preserves the same selection rule as the main text while exposing the complete $k \in \{5, 10, 20\}$ breakdown.

This table makes it possible to verify whether the observed trends persist beyond the compact main-text summary under the same selection rule.



(a) Reduced, shuffled, flat, and random-group cue variants

(b) Scope swap, identical-scope, and layout variants

(c) Stage-placement, ordering, and router variants

Figure 7: Extended structural ablations. (a) Semantic cue and family-structure variants. (b) Stage-semantics and layout variants. (c) Stage-placement, ordering, and router variants.

C Extended Structural Ablations

Figure 7 expands the structural study in Figure 4 with the broader sweep of cue variants, stage-semantics controls, family-grouping comparisons, and deeper layout comparisons that are too detailed for the main text.

Panel (a) tests whether semantic grouping matters beyond a flat bag of scalars. Panel (b) examines whether the chosen macro, mid, and micro roles matter beyond merely having three blocks. Panel (c) reports the broader family of stage-placement and router variants from which the final design was selected.

D Protocol Sensitivity and Supplementary Views

Table 8 records the supplementary checks that are most likely to arise once the main protocol is understood: whether the seen-target focus materially changes qualitative conclusions, and whether minor protocol adjustments overturn the main ranking pattern.

Table 6: Bounded candidate grids used in the final paper track.

(a) Shared Optimization Pool

Parameter	Candidate sets
Learning rate	Dataset-specific log-uniform interval; see panel (d)
Weight decay	$\{5 \times 10^{-7}, 10^{-6}, 10^{-5}, 5 \times 10^{-5}, 10^{-4}, 1.5 \times 10^{-4}\}$
Max history length	$\{10, 20, 30\}$ or $\{10, 20, 30, 50\}$
Width / embedding	$\{64, 96, 112, 128, 160\}$ or $\{96, 112, 128, 160, 192\}$
Inner width	$\{128, 192, 224, 256, 320\}$ or $\{192, 224, 256, 320, 384\}$
Layers / heads	Layers $\{1, 2, 3, 4\}$; heads $\{1, 2, 4, 8\}$ if applicable
Dropout	Hidden / ratio: $\{0.10, 0.12, 0.13, 0.15, 0.18\}$ Recurrent: $\{0.10, 0.15, 0.20, 0.25, 0.30\}$ Attention: $\{0.06, 0.08, 0.10, 0.12, 0.15\}$ or $\{0.08, 0.10, 0.12, 0.15, 0.20\}$

(b) Baseline-Family Additions

Model(s)	Additional candidate sets
TiSASRec	Time span $\{64, 128, 256, 384, 512\}$
GRU4Rec	Recurrent dropout $\{0.10, 0.15, 0.20, 0.25, 0.30\}$
DuoRec / FEARec	$\tau\{0.16, 0.18, 0.20, 0.22, 0.24\}$ Contrastive weight $\{0.02, 0.03, 0.04, 0.05, 0.06\}$ Semantic weight $\{0, 0.04, 0.05, 0.08, 0.10\}$ or $\{0.04, 0.05, 0.08, 0.10, 0.12\}$
BSARec	$\alpha\{0.35, 0.50, 0.55, 0.70\}$ $c\{2, 3, 5, 7\}$
DIF-SR / FDSA	Attribute width $\{96, 128, 160, 192\}$ λ_{attr} $\{0.08, 0.09, 0.10, 0.12, 0.14\}$ or $\{0.09, 0.10, 0.12, 0.14, 0.15\}$
DIF-SR	Fusion $\{\text{gate}, \text{sum}, \text{concat}\}$
FAME	Expert count $\{2, 3, 4, 5, 6\}$

(c) RouteRec-Specific Additions

Parameter	Candidate sets
Backbone depth / dropout	Depth: $\{1, 2, 3\}$ Dropout: $\{0.10, 0.12, 0.14, 0.15, 0.16, 0.17\}$ $\{0.18, 0.19, 0.20, 0.22, 0.24\}$
Weight decay	$\{3.125, 5, 6, 10, 15, 20, 50\} \times 10^{-7}$
Routing capacity	Expert scale $\{2, 3, 4\}$ or $\{2, 3, 4, 5\}$ for KuaiRec router width $\{32, 64, 96, 128\}$
Routing dropout	Feature $\{0, 0.03, 0.05, 0.10\}$ attention $\{0.05, 0.06, 0.08, 0.10, 0.12\}$
Regularization	$\lambda_{\text{cons}}\{0, 2.5, 5, 8, 12\} \times 10^{-4}$ $\lambda_z\{0, 5, 10, 20\} \times 10^{-5}$
Anchor family	Bank size $\{10, 12, 16, 24\}$ internal width preset per family

(d) Dataset-Specific Learning-Rate Intervals

Dataset	Unified log-uniform LR interval
Beauty	1.5e-4 ∼ 2e-3
Foursquare	1.5e-4 ∼ 2.2e-3
KuaiRec	3e-4 ∼ 5e-3
LastFM	8e-5 ∼ 1.2e-3
ML-1M	1.5e-4 ∼ 2.2e-3
Retail Rocket	1.5e-4 ∼ 2.2e-3

Table 8: Protocol sensitivity and supplementary evaluation views.

Check	RouteRec	Strongest baseline	Observation
Seen-target MRR@20	–	–	Main comparison in the paper
Overall MRR@20	–	–	Qualitative ranking remains similar
Unseen-target accounting	–	–	Supplementary only
Alternate threshold check	–	–	Representative-dataset robustness
Alternate held-out cleanup	–	–	No reversal of the main conclusion

The scope of this table is intentionally narrow. Its purpose is to show that the qualitative conclusion does not hinge on a single fragile reporting view.

E Objective and Regularization Variants

Table 9 isolates the contribution of consistency, z-loss, and balance regularization using ranking quality, route consistency, and stability indicators. This analysis justifies the final objective and helps prevent the main-text routing diagnostics from being over-interpreted as a trivial consequence of the consistency regularizer alone.

Table 9: Objective and regularization variants.

Variant	MRR@20	Consistency	Stability
No auxiliary loss	–	–	–
KNN + consistency	–	–	–
z-loss only	–	–	–
Balance loss only	–	–	–
Consistency + z-loss	–	–	–
Full objective	–	–	–

This compact table shows whether the full objective improves MRR@20 while preserving routing consistency and stability, and whether the diagnostic notion of consistency remains informative once the auxiliary loss is removed.

F Efficiency and Model Cost

Table 10 provides a compact cost summary for the claim that the routing branch is lightweight. Rather than turning the paper into a systems study, this section reports the rough parameter and runtime overhead for all compared methods under the shared protocol.

A compact cost comparison is sufficient here to anchor the lightweight claim and to show that the observed gains are not being attributed solely to an unchecked increase in computation.

Table 7: Full seen-target experimental results across all cutoffs. Datasets marked with † are sampled processed subsets. Avg. Rank is computed over all rows in this table, where lower is better. The RouteRec column is shaded for readability.

Dataset	Metric	SASRec	GRU4Rec	TiSASRec	FEARec	DuoRec	BSARec	FAME	DIF-SR	FDSA	RouteRec
Beauty	HR@5	0.0974	0.0372	0.0974	0.0974	0.0974	0.0430	0.0258	0.0802	0.0946	0.1054
	HR@10	0.1347	0.0716	0.1318	0.1261	0.1146	0.0516	0.0315	0.1060	0.1347	0.1616
	HR@20	0.1719	0.1032	0.1719	0.1719	0.1719	0.0602	0.0516	0.1433	0.1719	0.2014
	NDCG@5	0.0629	0.0227	0.0765	0.0603	0.0625	0.0289	0.0192	0.0602	0.0779	0.0803
	NDCG@10	0.0748	0.0340	0.0878	0.0696	0.0680	0.0316	0.0211	0.0685	0.0909	0.0982
	NDCG@20	0.0842	0.0418	0.0980	0.0810	0.0827	0.0339	0.0260	0.0778	0.1003	0.1083
	MRR@5	0.0514	0.0179	0.0694	0.0476	0.0506	0.0242	0.0170	0.0533	0.0725	0.0721
	MRR@10	0.0563	0.0226	0.0742	0.0515	0.0528	0.0253	0.0178	0.0567	0.0779	0.0794
	MRR@20	0.0588	0.0247	0.0770	0.0544	0.0569	0.0260	0.0190	0.0592	0.0805	0.0822
Retail Rocket	HR@5	0.4869	0.4475	0.4700	0.4647	0.4611	0.4413	0.4365	0.4305	0.4344	0.4883
	HR@10	0.5605	0.5211	0.5435	0.5441	0.5377	0.4703	0.4697	0.4365	0.5133	0.5612
	HR@20	0.6316	0.5863	0.6135	0.6200	0.6134	0.4986	0.5003	0.4409	0.5886	0.6313
	NDCG@5	0.3860	0.3620	0.3724	0.3678	0.3616	0.3869	0.3826	0.3881	0.3396	0.3914
	NDCG@10	0.4099	0.3858	0.3963	0.3935	0.3864	0.3963	0.3934	0.3900	0.3651	0.4150
	NDCG@20	0.4280	0.4023	0.4139	0.4127	0.4057	0.4034	0.4012	0.3911	0.3842	0.4327
	MRR@5	0.3524	0.3335	0.3399	0.3355	0.3284	0.3685	0.3644	0.3733	0.3082	0.3591
	MRR@10	0.3623	0.3434	0.3498	0.3462	0.3388	0.3724	0.3689	0.3741	0.3187	0.3688
	MRR@20	0.3673	0.3479	0.3547	0.3515	0.3441	0.3743	0.3710	0.3744	0.3239	0.3737
Foursquare	HR@5	0.2518	0.1743	0.2474	0.2418	0.2474	0.2014	0.2014	0.2426	0.2478	0.2562
	HR@10	0.3233	0.2130	0.3157	0.3042	0.3197	0.2462	0.2546	0.3034	0.2998	0.3225
	HR@20	0.3833	0.2590	0.3721	0.3645	0.3717	0.2866	0.3034	0.3521	0.3477	0.3777
	NDCG@5	0.1782	0.1331	0.1793	0.1721	0.1721	0.1496	0.1475	0.1786	0.1834	0.1827
	NDCG@10	0.2014	0.1455	0.2016	0.1925	0.1955	0.1639	0.1646	0.1984	0.2001	0.2041
	NDCG@20	0.2167	0.1572	0.2159	0.2077	0.2088	0.1741	0.1769	0.2107	0.2123	0.2181
	MRR@5	0.1540	0.1195	0.1568	0.1492	0.1473	0.1324	0.1297	0.1574	0.1620	0.1584
	MRR@10	0.1636	0.1246	0.1660	0.1578	0.1569	0.1382	0.1368	0.1656	0.1689	0.1673
	MRR@20	0.1679	0.1278	0.1700	0.1619	0.1607	0.1410	0.1401	0.1690	0.1723	0.1712
ML-1M	HR@5	0.0957	0.0906	0.1018	0.0883	0.0990	0.1050	0.0976	0.0888	0.1046	0.0939
	HR@10	0.1478	0.1413	0.1538	0.1487	0.1520	0.1589	0.1557	0.1501	0.1594	0.1561
	HR@20	0.2254	0.2119	0.2375	0.2277	0.2305	0.2282	0.2212	0.2300	0.2430	0.2337
	NDCG@5	0.0605	0.0581	0.0677	0.0531	0.0640	0.0680	0.0667	0.0566	0.0656	0.0572
	NDCG@10	0.0770	0.0742	0.0844	0.0724	0.0811	0.0854	0.0853	0.0764	0.0835	0.0771
	NDCG@20	0.0965	0.0919	0.1054	0.0924	0.1009	0.1028	0.1018	0.0963	0.1043	0.0964
	MRR@5	0.0489	0.0474	0.0565	0.0416	0.0525	0.0559	0.0566	0.0460	0.0529	0.0452
	MRR@10	0.0556	0.0539	0.0633	0.0495	0.0595	0.0631	0.0642	0.0542	0.0603	0.0534
	MRR@20	0.0608	0.0587	0.0690	0.0550	0.0649	0.0678	0.0687	0.0595	0.0658	0.0585
LastFM†	HR@5	0.3638	0.2829	0.3522	0.3515	0.3526	0.3243	0.3097	0.3362	0.3430	0.3562
	HR@10	0.3994	0.3090	0.3780	0.3765	0.3831	0.3453	0.3312	0.3617	0.3657	0.3794
	HR@20	0.4252	0.3341	0.4038	0.3962	0.4089	0.3653	0.3508	0.3863	0.3909	0.4070
	NDCG@5	0.3152	0.2589	0.3112	0.3086	0.2984	0.2956	0.2832	0.3067	0.3097	0.3178
	NDCG@10	0.3267	0.2673	0.3196	0.3167	0.3083	0.3024	0.2900	0.3149	0.3171	0.3253
	NDCG@20	0.3332	0.2736	0.3260	0.3216	0.3149	0.3075	0.2950	0.3212	0.3234	0.3323
	MRR@5	0.2988	0.2509	0.2976	0.2942	0.2803	0.2860	0.2742	0.2969	0.2986	0.3049
	MRR@10	0.3037	0.2543	0.3011	0.2976	0.2844	0.2888	0.2770	0.3002	0.3016	0.3079
	MRR@20	0.3054	0.2561	0.3027	0.2989	0.2863	0.2902	0.2784	0.3020	0.3033	0.3099
KuaiRec†	HR@5	0.0945	0.0842	0.0951	0.0951	0.0940	0.0902	0.0902	0.0891	0.0907	0.1000
	HR@10	0.1055	0.0945	0.1011	0.1005	0.0995	0.0918	0.1011	0.0918	0.0967	0.1060
	HR@20	0.1235	0.1109	0.1109	0.1158	0.1109	0.0923	0.1186	0.0995	0.1109	0.1235
	NDCG@5	0.0909	0.0756	0.0897	0.0902	0.0902	0.0885	0.0881	0.0863	0.0845	0.0946
	NDCG@10	0.0943	0.0790	0.0916	0.0919	0.0919	0.0890	0.0915	0.0871	0.0865	0.0965
	NDCG@20	0.0988	0.0831	0.0941	0.0958	0.0948	0.0892	0.0959	0.0891	0.0901	0.1009
	MRR@5	0.0897	0.0728	0.0879	0.0886	0.0890	0.0879	0.0874	0.0853	0.0824	0.0928
	MRR@10	0.0910	0.0742	0.0887	0.0893	0.0897	0.0882	0.0888	0.0857	0.0833	0.0936
	MRR@20	0.0922	0.0753	0.0894	0.0903	0.0905	0.0882	0.0899	0.0863	0.0843	0.0948
Avg. Rank↓		3.45	8.99	3.71	5.92	5.52	6.56	7.03	6.40	5.02	2.40

G Extended Diagnostics

Figure 8 groups the main routing diagnostics into one coordinated layout so that stage usage, sharpness, and semantic consistency can be read together.

These diagnostics are organized from coarse structure to semantic interpretation. The first two panels show how broad

Table 10: Model cost and efficiency summary for all compared methods.

Model	Params	Active params	Train-time ratio	Infer-time ratio
GRU4Rec	–	–	–	–
SASRec	–	–	–	–
TiSASRec	–	–	–	–
DuoRec	–	–	–	–
FEARec	–	–	–	–
BSARec	–	–	–	–
FAME	–	–	–	–
DIF-SR	–	–	–	–
FDSA	–	–	–	–
RouteRec	–	–	–	–

(a) Expert-usage heatmaps by stage

(b) Entropy and effective-expert-count summary

(c) Stage-wise route consistency

(d) Feature-bucket route patterns

Figure 8: Routing diagnostics. (a) Expert usage by stage. (b) Entropy and effective expert count. (c) Stage-wise route consistency. (d) Feature-bucket routing patterns.

or sharp routing is at each stage, while the last two panels connect those patterns back to feature similarity and interpretable behavioral regimes.

H Behavior-Slice Analysis

Because RouteRec is motivated by heterogeneous behavior, a slice-wise view is a natural complement to the aggregate metrics. This appendix section first defines the diagnostic slices explicitly, then expands the main-text regime analysis with per-dataset breakdowns, additional slices, and more detailed routing summaries.

These slices are used only for post hoc diagnosis. They are computed from simple observable statistics of the interaction history, not from optimized supervision targets, and not from labels learned jointly with the model.

Table 11: Diagnostic slice definitions used in the behavior-regime analysis.

Slice	Definition rule
Repeat-heavy	Prefix repeat ratio above a fixed percentile threshold
Fast-tempo	Mean inter-event gap below a fixed percentile threshold
Narrow-focus	Category or item-transition entropy below a fixed percentile threshold
Exploration-heavy	Novel-item or low-repeat ratio above a fixed percentile threshold

(a) Slice-wise MRR@20 across behavioral regimes

(b) Relative RouteRec gain and route concentration by slice

Figure 9: Extended behavior-regime analysis. (a) Slice-wise ranking quality. (b) Relative gain and routing concentration by slice.

Together, the two panels expose dataset-level heterogeneity that would be too detailed for the main paper while making the slice construction explicit enough to avoid circular interpretation.

H.1 Qualitative Routing Cases

Table 12 provides a compact case-based view of the routing behavior for a small number of representative sessions. The goal is not to replace the aggregate analyses, but to make

(a) Low-resource transfer curves

(b) Frozen/finetuned/anchor and group/full router transfer

(c) Multi-source sequential order effect

Figure 10: Transfer variants. (a) Low-resource transfer curves. (b) Frozen, finetuned, anchor, and router-granularity comparisons. (c) Multi-source order effects.

the macro/mid/micro interpretation readable at the level of individual examples.

This table complements the summary statistics with directly inspectable routing examples.

I Extended Transfer Results

This section reports additional transfer experiments for low-resource and cross-domain adaptation.

The matrix provides the exact source-target view that is too detailed for the main paper. It is useful for identifying asymmetric transfer pairs and for checking whether router reuse is most effective between domains with similar temporal and repetition structure. For compactness, the table uses the abbreviations Bty, Fsq, KRec, LFM, ML1M, and RR.

Together, the table and figure connect exact pairwise results to higher-level transfer patterns. The overall pattern suggests that transfer is asymmetric, most helpful in related domains and low-resource regimes, and partially preserved even under coarser router reuse.

Table 12: Representative qualitative routing cases.

Case	Prefix summary	Dominant routing pattern	Prediction outcome	Interpretation
Repeat-heavy session	–	–	–	Macro and micro routes reinforce revisit behavior
Fast-tempo exploratory session	–	–	–	Micro routing sharpens under short-horizon volatility
Narrow-focus session	–	–	–	Mid-stage routing concentrates on coherent intent
Preference-stable session	–	–	–	Gains narrow when routing demand is weak